Project 9: Fraud Detection in Financial Transactions

Objective:

To detect fraudulent transactions using anomaly detection techniques, reducing financial losses and enhancing customer trust. This system will identify suspicious activities in real-time to enable quick action and prevent fraud.

Detailed Plan:

1. Data Collection:
    - Gather transactional data, including:
        - Timestamps
        - Transaction amounts
        - User IDs and account details
        - Transaction types (e.g., online, in-store)
        - Geolocation and device information
    - Source data from financial institutions, payment gateways, or simulated datasets.
    - Ensure data privacy and compliance with regulations like GDPR.

2. Data Preprocessing:
    - Handle missing or incomplete data using imputation techniques.
    - Normalize numerical features, such as transaction amounts, for uniformity.
    - Encode categorical features like transaction types and user demographics.
    - Detect and handle data imbalances:
        - Use oversampling techniques like SMOTE (Synthetic Minority Oversampling Technique).